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RED TEAMER

## **Brief Research Report: Fields in Red Teaming**

### **1. Network Penetration Testing**

Focuses on attacking internal and external networks by exploiting misconfigurations, weak services, exposed ports, and insecure protocols. Techniques include port scanning, service enumeration, and lateral movement.

**Objective:** Gain unauthorized network access and pivot across systems.

### **2. Web Application Red Teaming**

Targets web applications using vulnerabilities such as SQL Injection, XSS, CSRF, SSTI, and logic flaws. Emphasizes chaining vulnerabilities for full compromise.

**Objective:** Compromise web apps and underlying servers.

### **3. Active Directory (AD) Attacks**

Involves attacking Windows domain environments using techniques like Kerberoasting, Pass-the-Hash, Golden Tickets, and privilege escalation.

**Objective:** Achieve domain admin-level control.

### **4. Phishing & Social Engineering**

Simulates human-targeted attacks via phishing emails, SMS (smishing), voice calls (vishing), and fake login portals.

**Objective:** Bypass technical controls by exploiting human behavior.

## 5. Malware Development

Designs custom payloads such as trojans, loaders, droppers, and backdoors to evade antivirus and EDR solutions.

**Objective:** Maintain stealthy persistence and command execution.

## 6. Command & Control (C2) Infrastructure

Builds attacker-controlled infrastructure for remote communication with compromised systems using HTTP(S), DNS, or covert channels.

**Objective:** Sustain long-term access without detection.

## 7. Post-Exploitation Operations

Performed after initial access; includes credential dumping, privilege escalation, lateral movement, and data exfiltration.

**Objective:** Maximize impact after compromise.

## 8. Cloud Red Teaming

Targets cloud platforms (AWS, Azure, GCP) by abusing IAM roles, misconfigured storage, exposed APIs, and weak secrets.

**Objective:** Compromise cloud resources and services.

## 9. Wireless Attacks

Focuses on Wi-Fi, Bluetooth, and RF attacks such as Evil Twin, deauthentication, and rogue access points.

**Objective:** Gain network access wirelessly.

## 10. Physical Security Testing

Tests real-world access controls including badge cloning, tailgating, lock picking, and hardware implant placement.

**Objective:** Assess physical security resilience.

## 11. Red Team Automation

Uses frameworks and scripts to automate attack chains (e.g., Caldera, Metasploit, custom Python tools).

**Objective:** Increase attack efficiency and repeatability.

## 12. Evasion Techniques

Specializes in bypassing antivirus, EDR, SIEM, sandboxing, and behavior-based detection.

**Objective:** Remain undetected during operations.

## 13. OSINT (Open-Source Intelligence)

Collects publicly available information from social media, leaked databases, job postings, and metadata.

**Objective:** Prepare realistic and targeted attacks.

#### **14. API & Microservices Attacks**

Targets REST/GraphQL APIs, JWT tokens, broken authentication, rate limits, and insecure integrations.

**Objective:** Exploit backend logic and data exposure.

#### **15. Mobile Application Red Teaming**

Focuses on Android and iOS apps, reverse engineering APK/IPA files, insecure storage, and API abuse.

**Objective:** Compromise mobile ecosystems.

#### **16. Insider Threat Simulation**

Simulates attacks from a trusted internal user abusing legitimate access.

**Objective:** Test internal monitoring and privilege boundaries.

#### **17. Supply Chain Attacks**

Targets third-party vendors, CI/CD pipelines, libraries, and update mechanisms.

**Objective:** Compromise targets indirectly via trusted sources.

#### **18. Purple Teaming**

A collaborative field combining Red Team (attack) and Blue Team (defense) to improve detection and response.

**Objective:** Strengthen overall security posture.